

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Pre-Tutorial Assignment 6: Solutions

1. Write the expression for the voltage $v_c(t)$ across the capacitor C for $t > 0$ (Figure 1). The numerical values of the circuit elements are as follows: $R_1 = 2\Omega$, $R_2 = 4\Omega$, $C = 10\mu F$ and $V_0 = 9V$.

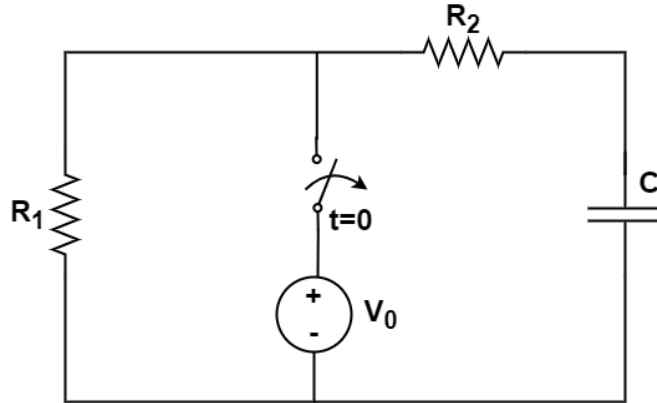


Figure 1: Electric circuit with an independent voltage source, a switch, resistors and a capacitor. The switch opens at $t = 0$.

Solution

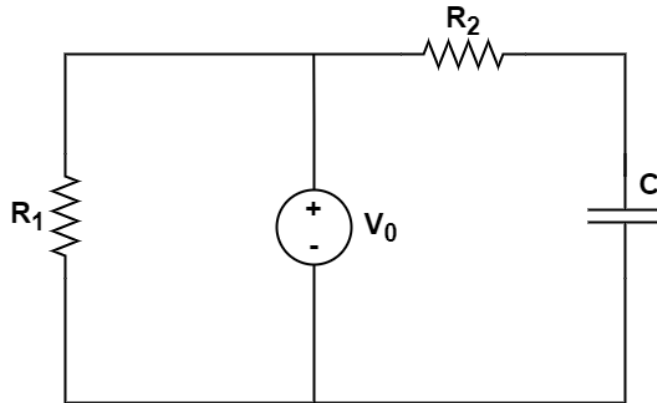


Figure 2: The circuit at $t = 0^-$.

The switch is closed at $t = 0^-$ and the corresponding circuit is shown in Figure 2. The capacitor is fully charged to $v_c(0^-) = V_0 = 9V$.

The switch opens at $t = 0$ and the corresponding circuit for $t > 0$ is shown in Figure 3. The capacitor starts discharging. Also, $v_c(0^+) = v_c(0^-)$. We see a series RC circuit with a time-constant $\tau_{RC} = (R_1 + R_2)C = (2 + 4) \times 10 \times 10^{-6}s = 60\mu sec$. Thus, the expression for $v_c(t)$ for $t > 0$ can be written as follows.

$$v_c(t) = v_c(0^+)e^{-\frac{t}{\tau_{RC}}} = 9e^{-\frac{t}{60 \times 10^{-6}}}(1)$$

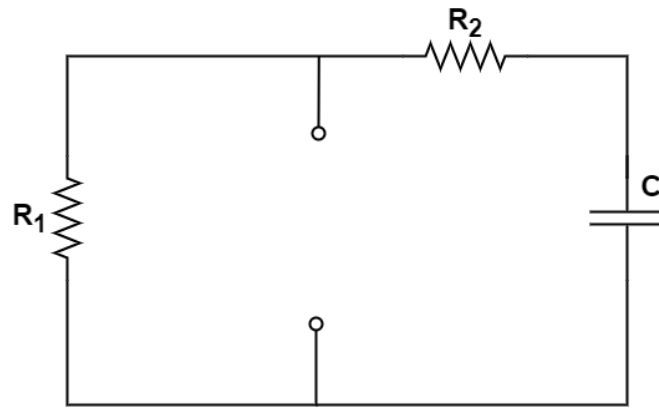


Figure 3: The switch opens at $t = 0$. The circuit is shown for $t > 0$.

2. Construct the following devices using a positive edge triggered JK Flip-Flop and logic gates.
 - (a) A positive edge triggered D Flip-Flop.
 - (b) A negative edge triggered T Flip-Flop.

Solution

The circuits implementing both the required devices are shown below:

